

AVYAKT GARG

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EDUCATION

M.S. in Computer Science
University of Wisconsin Madison

Sep 2025 - May 2027
GPA : 4/4

B.E. in Computer Science and M.Sc. in Biological Sciences
Birla Institute of Technology and Science, Pilani

Nov 2020 - May 2025
GPA : **9.01/10**

RELEVANT COURSES

Data Structures & Algorithms, Machine Learning, Object Oriented Programming, Artificial Intelligence, Database Systems, Computer Networks, Operating Systems, Microprocessors, Statistics, Human Computer Interaction, Computational Logic

PROFESSIONAL EXPERIENCE

Uber | *Software Intern*

Jul 2024 - Jun 2025

- Developed an upcoming knowledge work marketplace for **5+** countries focusing on data annotation and ML model training
- Implemented a rate-card system in **Java SpringBoot** framework using MVC and developed **gRPC** APIs for service integration
- Created a **Kafka** based notification engine with optimized cadence scheduling, enhancing customer retention funnel by **3X**
- Built a debug tool with 7 automated checks for **Uber AI Solutions**, reducing developer dependency on recurring issues by **80%**

University of Saskatchewan | *Mitacs Research Intern*

May 2023 - Aug 2023

- Collaborated with [Prof. Jaswant Singh](#) from **biomedical sciences**, focusing on data analysis for cattle health and reproduction
- Automated the daily weighing of cattle via developing RFID tags based data collection and analysis system saving **3K+** hrs/year
- Revamped the cattle management database with over **6600+** entries for **90%** faster queries via establishing genealogy links

Indian Institute of Technology, Ropar | *Research Intern*

Jul 2022 - Sept 2022

- Enabled cattle **health monitoring** by training and evaluating various **ML** models using data from farm based **IoT sensors**
- Analyzed **4k+** data points for cattle behavior prediction including movement patterns, estrus detection to aid farming operations

MapmyIndia | *Software Intern*

May 2022 - July 2022

- Created a web application for delivery executives based on the **traveling salesman algorithm** to find the most optimal trip path
- Integrated an interactive map API with **HTML**, **CSS** and **JavaScript** with features like resizing and custom marker placement

PROJECTS

Graph Learning for Disease Prediction | *Graduate Project*

- Engineered **Patient Similarity Networks** and **Bipartite Patient-Attribute Graphs** to model non-i.i.d. clinical relations.
- Developed **GraphSAGE** and **Node2Vec** pipelines, achieving 83.5% accuracy and 0.913 AUC-ROC with Nested Cross-Validation.
- Integrated GNNExplainer for clinical transparency and deployed a risk assessment web app using **FastAPI** and **Streamlit**.

Patterning Protein Localisation in Endothelial Cells | *Prof. Syamantak Majumder*

- Spearheaded development of an automated cell segmentation software to detect protein localization in differently stained images.
- Used **Otsu's thresholding** in python with **OpenCV**, saving 6+ biologist hrs/study while improving nuclear detection by 2X

Deep Learning Framework for prediction of Intracranial Pressure (ICP) using OCT Scans | *Prof. S. Raman*

- Built a 3D ResNet-18 classifier for intracranial pressure prediction using Optical Coherence Tomography(OCT) retinal scans
- Preprocessed 512-frame videos to isolate key regions of interest and fine-tuned architecture for safe patient threshold prediction

Machine Learning Techniques for Diagnosis of Alzheimer's Disease | *Prof. Bharat Richhariya*

- Achieved **96.88%** accuracy using ML models on 1000+ preprocessed 1.5T MRI scans from ADNI dataset with CAT-12 software
- Utilized **transfer learning** to save computational resources and improve the performance of the **3D Resnet 18** model

E-commerce Management System | *Coursework Project*

- Developed a **MySQL**-based user-friendly **e-commerce** platform which allowed the user to browse products and place orders.
- Designed a **Python Tkinter** based GUI which enables customers to track order status, view payment history and read feedback

ACADEMIC ACHIEVEMENTS

Globalink Research Scholarship — MITACS, Canada: Selected out of 30k+ applicants for research internship

Feb 2023

Scholarship Awardee — AWaDH, Govt. of India: For AI work in agriculture domain

Apr 2022

Gold Medalist — Delhi Public School R.K. Puram: 9 years of academic excellence

May 2019

86th Rank — JSTSE — Directorate of Education, Delhi: Science merit scholarship

Feb 2017

TECHNICAL SKILLS

Programming Languages: Java, Python, C/C++, SQL, HTML, JavaScript

Framework & Library: SpringBoot, OpenCV, TensorFlow, PyTorch, Scikit-learn, GraphSAGE, Node2Vec, Pandas, NumPy

Software Development: MVC Architecture, Kafka, Mockito, FastAPI, Streamlit, gRPC, REST, Docker, Git, Agile, JIRA